Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/22/2005
PI Name:	Monk, Timothy H Ph.D., D.Sc.		
Project Title:	Performance and Sleep Consequences of Slam Shifts in Schedule		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	monkth@upmc.edu	**Fax:**	FY 412-246-5300
PI Organization Type:	UNIVERSITY	**Phone:**	412-246-6413
Organization Name:	University of Pittsburgh Medical Center		
PI Address 1:	Director, Human Chronobiology Program, Clinical Neuroscience		
PI Address 2:	3811 O'Hara Street		
PI Web Page:			
City:	Pittsburgh	**State:**	PA
Zip Code:	15213	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	05/15/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ04HF76G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project comprises a ground-based study of how human sleep, circadian rhythms and performance are affected by abrupt “slam shifts” as are encountered by astronauts on space station when the work-rest schedule needs to be changed (e.g., to accommodate a supply vehicle rendezvous). The study will additionally determine whether 90-minute naps in the workday following such shifts can be helpful. Comparisons will also be made with data from previous University of Pittsburgh studies (NAG9-1234) of repeated gradual phase shifts that are within NASA crew scheduling constraints (Appendix K). The experiment seeks to answer several questions about coping with slam shifts (i.e., abrupt shifts in work/rest schedule of 6h), including the following: 1) In the light levels and comparative social isolation typical of ISS, how long does the human Endogenous Circadian Pacemaker (ECP) take (i.e. how many days) to entrain completely (at the correct phase angle) to: a) a 6h phase advance (slam shift out)? and b) a 6h phase delay (slam shift back)? 2) To what degree, and for how many nights, is objectively and subjectively recorded sleep disrupted by: a) a 6h phase advance		

	(slam shift out)? and b) a 6h phase delay (slam shift back)? 3) To what degree, and for how many days, is objectively measured performance and subjectively reported alertness and mood disrupted by a) a 6h phase advance (slam shift out)? and b) a 6h phase delay (slam shift back)? 4) To what extent (if any) are the disruptions noted in #2 and #3 ameliorated by a 90-minute daytime nap scheduled for the workday following each of the two slam shifts? 5) To what extent (if any) are the total disruptions in circadian rhythms, sleep, alertness and performance occasioned by slam shifts better or worse than those occasioned by the repeated gradual (appendix K - like) phase shifts we have studied in our current grant (NAG 9-1234)? 6) How will the disruptions noted in #3 vary by the particular task used?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	It is too early to present specific results from this study. A total of 42 subjects have been recruited and consented for the study. Of these 42 subjects, 18 have passed through or are currently part way through our rigorous screening procedures which include detailed physical and mental health screening interviews and tests as well as a night spent in our sleep lab with detailed polysomnographic measures taken. Of those subjects, four (m, 53y; m, 45y; 29y f; 20y m) have completed the entire 16-day protocol. One subject (f, 42y) started the protocol, but the study ended early due to the unexpected death of the subject's brother.
Bibliography Type:	Description: (Last Updated: 04/27/2011)
Articles in Peer-reviewed Journals	Monk TH, Buysse DJ, Billy BD, DeGrazia JM. "Using nine 2-h delays to achieve a 6-h advance disrupts sleep, alertness, and circadian rhythm." Aviat Space Environ Med. 2004 Dec;75(12):1049-57. PMID: 15619859 , Dec-2004
Articles in Peer-reviewed Journals	Monk TH. "The post-lunch dip in performance." Clin Sports Med. 2005 Apr;24(2):e15-23, xi-xii. PMID: 15892914 , Apr-2005
Articles in Peer-reviewed Journals	Mansour HA, Wood J, Chowdari KV, Dayal M, Thase ME, Kupfer DJ, Monk TH, Devlin B, Nimgaonkar VL. "Circadian phase variation in bipolar I disorder." Chronobiol Int. 2005;22(3):571-84.PMID: 16076655 , Jun-2005
Articles in Peer-reviewed Journals	Monk TH. "Aging human circadian rhythms: conventional wisdom may not always be right." J Biol Rhythms. 2005 Aug;20(4):366-74. Review. PMID: 16077155 , Aug-2005
Books/Book Chapters	Monk TH. "Sleep & sleep disorders." in "The encyclopedia of aging : a comprehensive resource in gerontology and geriatrics." Ed. R. Schulz. New York : Springer Pub. Company, c2006., Jan-2006
Presentation	Monk, T.H.; Buysse, D.J.; Billy, B.D.; DeGrazia, J.M.; Clark, M.L. "Performance Consequences of Repeated Phase Shifts within Appendix K Guidelines and Performance and Sleep Consequences of Slam Shifts in Schedule" 2005 NASA Bioastronautics Investigator Meeting Jan-2005